

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

C04: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning
(BL 5)

Assessment Tool Weightage: 10%

QUESTIONS: 15

Total Marks:20

Annexure A

Comparative Analysis

Code of Conduct:

I Abinesh H (192424387) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

<i>Criteria</i>	<i>Conceptual Understanding</i>	<i>Accuracy of Answers</i>	<i>Application of Knowledge</i>	<i>Completeness of Response</i>	<i>Clarity & Presentation</i>	<i>Total</i>
<i>Weightage</i>	30%	15%	20%	20%	15%	
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Signature of the student: Abinesh H

Total Marks :

Annexure A

Short Answer Test

1. Compare **Linear Regression and Polynomial Regression** for predicting housing prices. Analyze differences in model complexity, ability to capture non-linear relationships, risk of overfitting, and interpretability. Discuss scenarios where each model performs better.
2. Compare **Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM)** networks for time-series prediction tasks such as stock price forecasting. Analyze differences in handling long-term dependencies, vanishing gradient problem, computational complexity, and prediction accuracy.
3. Compare **Naïve Bayes and Support Vector Machines (SVM)** for classification problems such as sentiment analysis. Analyze differences in assumptions, decision boundaries, computational efficiency, and performance on high-dimensional data. Discuss scenarios where each method is preferable.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	15	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	20	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Linear Regression vs Polynomial Regression for Housing Price Prediction

Both **Linear Regression** and **Polynomial Regression** can be used to predict housing prices, but they differ in how they model relationships between features and price.

Factor	Linear Regression	Polynomial Regression
Relationship	Assumes a linear relationship	Captures non-linear relationships
Complexity	Simple	More complex
Model equation		
Overfitting	Lower risk	Higher risk, especially with high degree
Interpretability	Easy to interpret	More difficult
Training	Fast	Usually slower
Flexibility	Limited	More flexible
Feature engineering	Minimal	Polynomial features required

Linear Regression

Linear Regression works well when housing price changes approximately proportionally with factors such as **area, number of rooms, or location score**.

Example: If price consistently increases as house area increases, a linear model may perform well.

Polynomial Regression

Housing prices often have non-linear relationships. For example, the increase in price may become much larger for luxury homes. Polynomial Regression can capture such curves.

However, using a very high polynomial degree can make the model fit noise in the training data, resulting in **overfitting**.

When to Use

Linear Regression is preferable when:

- The relationship is approximately linear.
- The dataset is relatively simple.
- Interpretability is important.
- A fast and stable model is required.

Polynomial Regression is preferable when:

- The relationship between features and price is clearly non-linear.
- A curved relationship improves prediction.
- The polynomial degree is carefully controlled.

Conclusion: Start with Linear Regression as a baseline. If the data shows significant non-linear patterns, Polynomial Regression can provide better predictions, provided overfitting is controlled.

2. RNN vs LSTM for Time-Series Prediction

Both **Recurrent Neural Networks (RNNs)** and **Long Short-Term Memory (LSTM)** networks are designed for sequential data such as stock prices, weather measurements, and sales.

Factor	RNN	LSTM
Basic structure	Simple recurrent connections	Memory cells with gates
Short-term dependencies	Good	Good
Long-term dependencies	Limited	Excellent
Vanishing gradient	More affected	Significantly reduced
Computational complexity	Lower	Higher
Training speed	Faster	Slower
Memory mechanism	Basic hidden state	Cell state + gates
Accuracy on long sequences	Often lower	Usually better

RNN

An RNN passes information from previous time steps to later time steps. It works effectively when recent observations are more important.

However, during training, gradients can become extremely small as they are propagated through many time steps. This is known as the **vanishing gradient problem**, making it difficult for a basic RNN to learn long-term dependencies.

LSTM

LSTM addresses this problem using three main gates:

- **Forget gate** – decides what information to discard.
- **Input gate** – decides what new information to store.
- **Output gate** – decides what information to output.

This allows LSTM to retain important information over longer sequences.

Stock Price Prediction

For stock forecasting, past information may influence future behavior over different time scales. LSTM can therefore be advantageous when meaningful long-term patterns exist.

However, stock prices are highly noisy and influenced by external factors, so **LSTM does not guarantee better financial predictions** simply because it is more complex.

When to Use

RNN is preferable when:

- The sequence is relatively short.
- Short-term dependencies dominate.
- Computational resources are limited.
- A simple recurrent model is sufficient.

LSTM is preferable when:

- Long-term dependencies are important.
- The sequence is long.
- The basic RNN suffers from vanishing gradients.
- Higher computational cost is acceptable.

Conclusion: LSTM is generally more suitable for complex long-sequence prediction because of its memory mechanism, while RNN is simpler and computationally cheaper.

3. Naïve Bayes vs SVM for Sentiment Analysis

Naïve Bayes (NB) and **Support Vector Machine (SVM)** are both effective classification algorithms for tasks such as sentiment analysis.

Factor	Naïve Bayes	SVM
Main assumption	Features are conditionally independent	Finds an optimal separating hyperplane
Decision boundary	Probabilistic	Margin-based
Training speed	Very fast	Usually slower
Prediction speed	Very fast	Fast after training
High-dimensional data	Excellent	Excellent
Small datasets	Often effective	Often effective
Interpretability	Relatively simple	Moderate
Probability output	Naturally available	Not inherently probabilistic
Text classification	Very suitable	Very suitable

Naïve Bayes

For sentiment analysis, a document can be represented using word frequencies or TF-IDF values.

Naïve Bayes estimates:

It assumes that features are conditionally independent given the class. This assumption is not completely true for natural language because words can depend on one another, but the method can still perform very well.

Advantages:

- Simple implementation
- Very fast training
- Works well with large vocabularies
- Requires relatively little training data
- Suitable for real-time classification

Support Vector Machine

SVM finds a decision boundary that maximizes the **margin between classes**.

For sentiment analysis, the input may contain thousands of word or n-gram features. SVM can handle this high-dimensional sparse representation effectively.

For example:

Positive reviews → one side of hyperplane

Negative reviews → other side of hyperplane

The closest training examples are called **support vectors**, and they influence the decision boundary.

When to Use

Naïve Bayes is preferable when:

- Fast training is required.
- The dataset is large and sparse.
- A simple baseline is needed.
- Computational resources are limited.
- Probabilistic class estimates are useful.

SVM is preferable when:

- Classification accuracy is the priority.
- The dataset contains high-dimensional text features.
- A strong decision boundary is needed.
- Training time is less important than classification performance.

Overall Comparison

For **sentiment analysis**, both methods can perform well. **Naïve Bayes** is usually simpler and faster, making it an excellent baseline. **SVM** can provide stronger classification performance when the text representation is high-dimensional and the classes are reasonably separable.

Conclusion:

Choose Naïve Bayes for **speed, simplicity, and efficient text classification**, while SVM is preferable when **robust classification boundaries and potentially higher predictive performance** are more important.